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DATA DESCRIPTOR

First-principles data for solid solution niobium-tantalum-vanadium alloys with body-centered-cubic structures

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We present four open-source datasets that provide results of density functional theory (DFT) calculations of ground-state properties of refractory solid solution binary alloys niobium-tantalum (NbTa), niobium-vanadium (NbV), tantalum-vanadium (TaV), and ternary alloys NbTaV ordered in body-centered-cubic (BCC) structures with 128 Bravais lattice sites. The first-principles code used to run the calculations is the Vienna Ab-Initio Simulation Package. The calculations have been collected by uniformly sampling chemical compositions across the entire compositional range. For each chemical composition, the calculations have been run for 100 randomized arrangements of the constituents on the BCC lattice sites. This sampling methodology resulted in running DFT simulations for a total of 3,100 randomized atomic configurations over 31 chemical compositions for each of the three binary alloys Nb-Ta, Nb-V, Ta-V, and a total of 10,500 randomized atomic structures over 105 chemical compositions for the ternary alloys Nb-Ta-V. For each atomic configuration, geometry optimization has been performed, and the data released contains information about each step of geometry optimization for each atomic configuration.

Background & Summary

In multiple principal element alloys (also called high entropy alloys, HEAs) systems, the Gibbs energy functions of the predominant solid solution phases often display multiple local minimums (saddle points), which create ample opportunities to form immiscibility gaps in these systems for phase decomposition^{1–4}. The natural way for developing new alloys is to search for uniform solid solutions. However, the presence of nanoscale size non-homogeneities often improves alloy properties⁵. The phase decomposition has an important role in forming heterogeneous nanoscale domains in microstructure, from which numerous interfaces between different nanodomains can be created and serve as barriers for dislocation movement, thereby improving strength for a wide range of temperatures. Recently, this strategy has been successful to develop radiation resistant alloys⁶.

The higher the decomposition temperature of the material, the more likely it can achieve high strength at higher temperature. Considerable research has been performed on the face-centered-cubic (FCC), equiatomic, quinary HEA chrome-manganese-iron-cobalt-nickel (CrMnFeCoNi) and its derivatives⁷ and interesting mechanical properties have been reported⁸. In contrast, fewer results have been reported for body-centered-cubic (BCC) HEAs, especially those containing refractory elements. Among the refractory HEAs (RHEAs), two equiatomic quinary stand out as model systems for investigation because they can be synthesized as single-phase solid solutions: vanadium-niobium-molybdenum-tantalum-tungsten (VNbMoTaW)⁹ and titanium-zirconium-hafnium-niobium-tantalum (TiZrHfNbTa)^{10,11}. These RHEAs and their derivatives have been proposed as potential high-temperature materials due to the expectation that their high melting points will result in high strengths at elevated temperatures. Indeed, compression tests to 1200 °C¹² have confirmed

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that the yield strength of the VNbMoTaW RHEA is superior to that of Inconel 718 above 800 °C. RHEAs such as VNbMoTaW and TiZrNbHfTa, with estimated melting temperatures (T_m) of 2700 °C (rule of mixtures) and 2000 °C (CALPHAD calculation), respectively, are being explored. However, the phase stability and phase separation in RHEAs is historically difficult to study because of the limitation of experimental capability at high temperatures and sluggish decomposition kinetics to reach equilibrium at low temperatures. For example, investigations into the phase stability of the Ti-Zr-Hf-Nb-Ta system have shown that TiZrNbHfTa has a single-phase BCC structure after 2 hours at 1000 °C and 3 hours at 1200 °C, but transformed into a BCC1+BCC2 structure with ZrHf-enriched and NbTa-enriched phases after 2 hours at 800 °C^{13,14}. However, additional work on TiZrNbHfTa by Schuh *et al.*, showed that after high-pressure torsion (HPT) at room temperature, the alloy consisted of a single BCC phase and subsequent annealing for 1 hour at 300 °C did not produce a phase transformation¹⁵. The phase separation at high temperature does not seem consistent with the single phase observed at low temperature.

Accurate prediction of formation energy for RHEAs using computational methods is of critical importance to accurately calculate the phase separation and microstructure evolution. This effort requires: (1) studying a crystal structure with a sufficiently large number of atoms to capture the effect of the configurational entropy, (2) reduce the boundary effect on the prediction of the formation energy, and (3) include effect of temperature. State-of-the-art density functional theory¹⁶ (DFT) based approaches used to compute the formation energy on large crystal structures are very expensive and typically only focus on static crystal structures at 0 Kelvin. This makes questionable the use of DFT based approaches to compute the formation energy for several atomic configurations that span multiple chemical compositions for several temperature. Additionally, the material space described by all the possible chemical composition and atomic disorders for the crystal structure of the high entropy alloy is high dimensional. The dimensionality of the material space increases exponentially with the number of pure elements mixed and the number of atoms in the crystal structure. Computation of ground-state structures of a binary alloy requires calculating the total energy for all possible configurations of placing atoms *A* and *B* on *N* sites of the underlying Bravais lattice. As the number of possible configurations 2^N is extremely high, even for relatively low values of *N*, it is computationally unfeasible to calculate the energy quantum mechanically for an exhaustive set of configurations even using state-of-the-art leadership class supercomputing facilities.

Present limitations of large-scale computational approaches are shedding new light on the potential benefit of integrating machine learning (ML) within traditional material science and quantum mechanics¹⁷. In particular, deep learning (DL) models have shown the ability to effectively capture relevant non-linearities triggered by the atomic configurations of an atomic system^{18–27}. Once trained, the trained DL model can be used for inference in a fraction of the time it would take to run a full DFT calculation, while still producing accurate results. This drastic reduction of time to predict properties of alloys using atomic information opens a promising path towards an effective acceleration of discovery and design of new multi-component alloys. The amount of data required to train DL models for the study of HEAs depends on the dimensionality of the chemical space to explore, the complexity of the DL architecture used, and the availability and quality of the data. In general, DL models benefit from large amounts of diverse and high-fidelity data. However, collecting such large amounts of high-fidelity data is challenging for HEAs. In particular, the design of HEAs often involves exploring a vast compositional space²⁸, which means that obtaining a comprehensive dataset covering the entire space is only possible by accessing and leveraging leadership class supercomputing facilities.

We present four new datasets that provide results of DFT calculations at 0 Kelvin for BCC structures with 128 Bravais lattice sites of solid solution binary alloys NbTa²⁹, NbV³⁰, TaV³¹, and solid solution ternary alloys NbTaV³². The first-principle code used to run the calculations is the Vienna Ab-Initio Simulation Package (VASP)^{33–37}. We carried out VASP calculations utilizing two DOE supercomputers: OLCF-Summit at Oak Ridge National Laboratory (ORNL) and NERSC-Perlmutter at the National Energy Research Scientific Computing Center (NERSC). The versions of the VASP software used are the version 6.3.2 on Summit and 6.2.1 on Perlmutter, both of which enable hardware acceleration on NVIDIA graphic processing units (GPUs). The calculations have been collected by uniformly sampling chemical compositions across the entire compositional range of the multi-component systems. The chemical compositions have been sampled starting from a pure metal and by progressively increasing the number of atoms of the other constituent by 4 for the binary alloys, and by progressively increasing the number of atoms of the other constituents by 8 for the ternary alloys. For each chemical composition of binaries and ternaries, 100 randomized atomic arrangements of the constituents on the BCC lattice sites have been sampled. The DFT simulations have been run for a total of 3,100 randomized atomic configurations over 31 chemical compositions for each of the three binary alloys NbTa, NbV, TaV, and for a total of 10,500 randomized atomic configurations over 105 chemical compositions for the ternary alloys NbTaV. Starting from an ideal BCC structure, geometry optimization has been performed on each atomic configuration until the geometry has been fully relaxed. This dataset will serve as starting point to compute the phonon contribution to the energy of the multi-component system NbTaV by taking the fully relaxed geometry for each atomic configuration and displacing the atoms from their position of equilibrium. In addition, using information from each step of geometry optimization from the output of the VASP calculations, this dataset also provides informative material to develop ML force fields. The four datasets for binary systems NbTa, NbV, TaV, and for the ternary systems NbTaV have been released open-source^{29–32}.

The remainder of this paper is structured as follows. Section “Methods” describes the input parameters used to run the VASP calculations. Section “Procedures for Data Generation” describes the process to generate the randomized atomic configurations and submit the calculations in batches to the job scheduler of the supercomputers for execution. Section “Technical Validation” comments on the results obtained from our calculations with respect to previous work performed on the same system with only 12 atoms in the BCC structure, as well as preliminary experimental low temperature observations provided by our experimental collaborators. Section “Usage Note” provides instructions on how to access and download the data. Section “Code Availability”

Software	Description	Version
VASP-6	Atomistic first principles package	6.3.2
ESSL	Numerical library	6.3.0
FFTW	Numerical fast Fourier transformation library	3.3.10
Spectrum MPI	MPI implementation	10.4.0.3
CUDA	NVIDIA GPU Toolkit	11.0
Python	High-level programming language	3.9
SQLite	Lightweight database engine	3.40

Table 1. Software Specification on OLCF Summit for the VASP6 calculations.

Software	Description	Version
VASP-6	Atomistic first principles package	6.2.1
Cray MPICH	MPI implementation	8.1.25
FFTW	Numerical fast Fourier transformation library	3.3.10
CUDA	NVIDIA GPU Toolkit	11.7
Python	High-level programming language	3.9
SQLite	Lightweight database engine	3.40

Table 2. Software Specification on NERSC-Perlmutter for the Workflow Components.

provides details on the open-source repository with the python scripts that we used to generate the randomized atomic configurations and the bash script that we used for job scheduling to submit the calculations to the supercomputers.

Methods

The electronic structures of alloys have been calculated using Vienna Ab initio Simulation Package (VASP)^{33–37}. Within this package the density functional theory (DFT)¹⁶ approach is used to describe the electron-electron interactions. This allows the reduction of the many-body Schrödinger equation to a set of single particle Kohn-Sham (KS) equations³⁸. The many particle electron interaction in DFT is captured by an universal exchange-correlation functional for which a wide choice of approximations have been proposed. In the present calculations, we employ the generalized gradient approximation within the Perdew-Burke-Ernzerhof³⁹ parametrization. The electron-ion interactions are described by pseudopotentials (PPs) using the plane-wave basis projector augmented-wave (PAW) approach⁴⁰. The PPs used in our calculations are the set provided with the respective VASP distributions. We chose PPs treating *s* and *p* semi-core states as valence states in the cases of V, Nb, and *p* semi-core states as valence states in case of Ta. These are the VASP-developer-recommended potentials for DFT calculations, as outlined in https://www.vasp.at/wiki/index.php/Available_PAW_potentials#Recommended_potentials_for_DFT_calculations. The electronic densities and potentials are expanded in plane-waves with an energy cutoff of 350 eV. A $2 \times 2 \times 2$ k-mesh and the VASP precision setting “normal” were used. Thus, such chosen mesh and cutoff energy guarantee accuracy of the calculation by dropping the error below 4 meV per atom. Methfessel-Paxton smearing of electronic occupations near the Fermi level were introduced with a broadening of 0.2 eV. The alloys are modeled by supercells containing 128 randomly distributed atoms. In the initial structure, the atoms occupy perfect BCC lattice sites. These initial structures were optimized until an energy change of less than 10^{-4} eV was achieved and forces acting on the atoms do not exceed 10^{-2} eV/Å.

Procedures for data generation

Creation of atomic structures and set-up of VASP6 calculations. The Python scripts that we used to generate the VASP6 cases to run are available at the ORNL-GitHub repository https://github.com/ORNLS/ScalableWorkflow_VASP_Calculations. The script `generate_vasp_atomic_configurations_binary.py` was used to generate atomic structures for the solid solution binary alloys Nb-Ta, Nb-V, and Ta-V. The variable `atomic_increment` was set to 4 atoms to sample 31 chemical compositions evenly distributed across the entire compositional range of the binary alloys. Similarly, the script `generate_vasp_atomic_configurations_ternary.py` was used to generate atomic structures for the solid solution ternary alloys Nb-Ta-V. The variable `atomic_increment` was set to 8 atoms to sample 105 chemical compositions evenly distributed across the entire compositional range of the ternary alloys. For the random generation of disordered atomic structures, we fixed the random seed to 42.

Both Python scripts generate a directory for each sampled chemical composition. Within each directory for a chemical composition, multiple sub-directories have been created, one per atomic configuration. For each sub-directory associated with an atomic configuration, a randomized atomic structure that mixes atoms of different constituents is generated by randomly shuffling an initial atomic structure provided in a POSCAR file using the function `numpy.random.shuffle()`. Once the execution of these python scripts is completed, each sub-directory associated with an atomic structure contains only the following files needed to start the VASP6 calculations: INCAR, POTCAR, POSCAR, KPOINTS. The file INCAR contains various parameters and

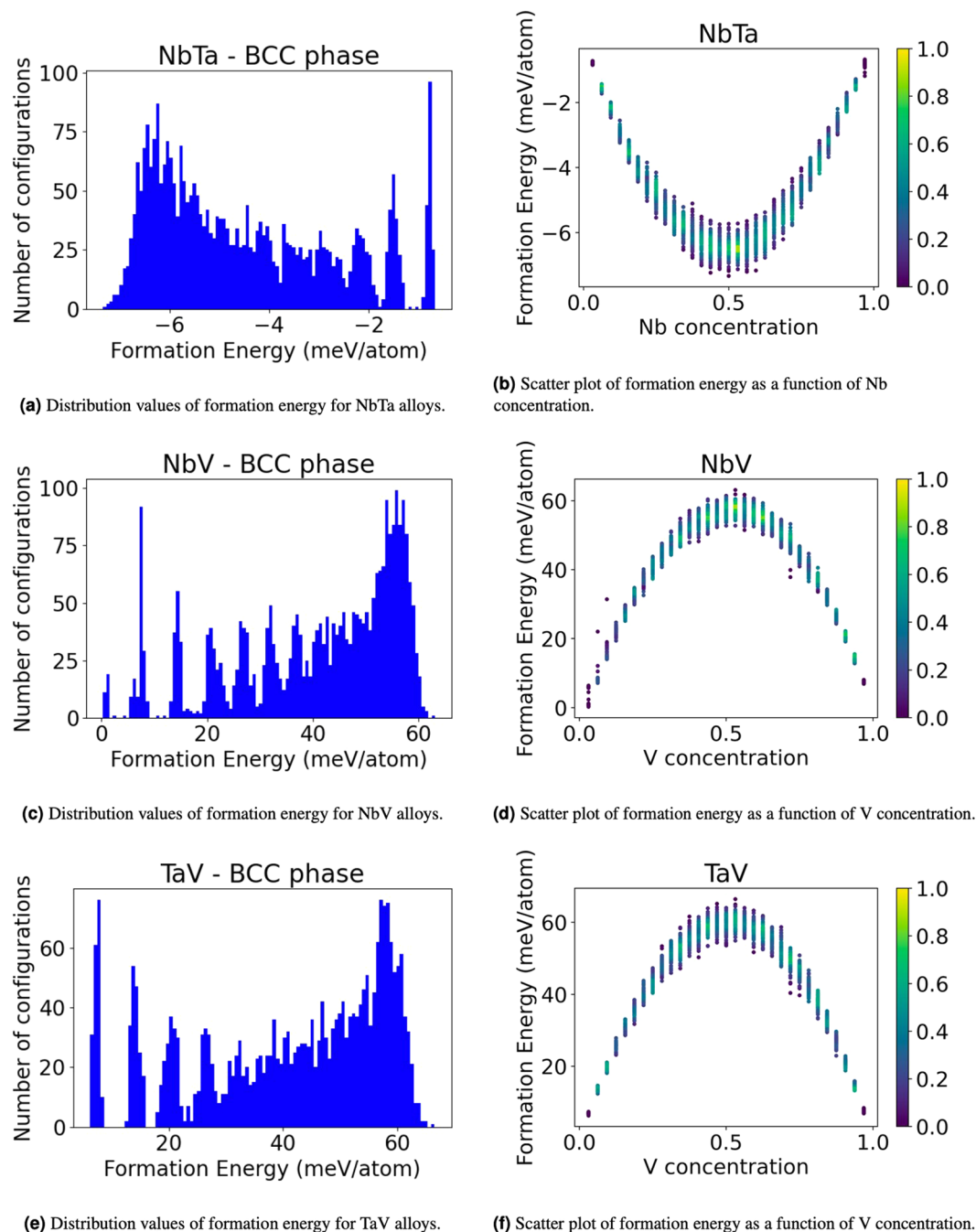


Fig. 1 Formation energy of solid solution NbTa, NbV, and TaV binary alloys with BCC structure.

settings for controlling the behavior of the electronic structure calculations, the file `POTCAR` contains information about the pseudopotential, the file `POSCAR` contains the information about the initial atomic structure before geometry optimization is performed, and the file `KPOINTS` contains the information about the quadrature points that need to be used for numerical integration in the complex plane.

Scalable scheduling of large ensemble of VASP6 calculations on US-DOE leadership class supercomputers. Our focus is on generating extensive and complete numerical data, aimed at enhancing future scientific research in the field of ML-assisted design of solid solution alloys with desired thermodynamic and mechanical functional properties. Our data collection process has been used to perform VASP calculations on a total of 19,800 atomic configurations across all binaries and ternaries, collectively amounting to approximately 120,000 compute node hours on OLCF-Summit and 100,000 node hours on NERSC-Perlmutter. For each configuration, we launched concurrent VASP computations using multiple GPUs distributed across multiple nodes on both supercomputing facilities.

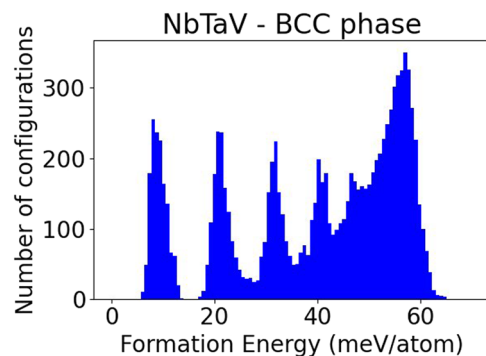


Fig. 2 Distribution values of formation energy for NbTaV alloys.

A significant challenge we faced was managing a vast quantity of DFT calculations that demands careful monitoring of tasks, identifying and tracking errors, and rerunning them when necessary, which is a tedious but crucial process. Drawing inspiration from the GNU parallel tool⁴¹, renowned for its user-friendly approach to executing multiple tasks concurrently, we have developed an in-house tool. This tool is designed for efficient and easy parallel execution and monitoring of multiple jobs. The cornerstone of our algorithm is its ability to maintain a task list in a lightweight database file supporting concurrent access by multiple processes. We employ SQLite3 for its lightweight nature, serverless architecture, cross-platform compatibility, and support with transactions for concurrent access by multiple processes.

By leveraging SQLite3's lock and unlock capabilities, which guarantee that only one process can modify a database file at a time, we ensure that multiple processes can pick up an unfinished job without causing duplications, while also maintaining records of completed tasks and any error status. This setup allows us to efficiently scale our resources; we can easily allocate more workers when additional compute time is available, or scale down according to the availability of compute nodes. Throughout this process, the global task list and status are maintained seamlessly in a database file, avoiding any conflicts.

Software specification to run VASP6 calculations on OLCF Summit. Part of the calculations were performed on the Summit computer system at the Oak Ridge Leadership computing facility. Summit is an IBM AC922 system which combines IBM Power9 CPUs with NVIDIA V100 GPUs. Table 1 shows the main software components and their versions used for this work that are installed on OLCF-Summit.

Software specification to run VASP6 calculations on NERSC Perlmutter. We used the Perlmutter supercomputer at the National Energy Research Scientific Computing Center (NERSC). Perlmutter is a Cray system which combines AMD Milan CPUs with NVIDIA A100 GPUs. Table 2 shows the software packages utilized in this project, including the primary software components and their respective versions used for this work that are installed on NERSC-Perlmutter.

Description of the Datasets

The datasets NbTa_BCC_SolidSolution_128atoms_VASP6²⁹, NbV_BCC_SolidSolution_128atoms_VASP6³⁰, TaV_BCC_SolidSolution_128atoms_VASP6³¹, and NbTaV_BCC_SolidSolution_128atoms_VASP6³² contain the input and output files of the VASP calculations for the binary alloys NbTa, NbV, TaV, and the ternary alloys NbTaV, respectively.

Each one of these datasets is structured with three hierarchical levels of directories. The highest level of each dataset is comprised of three folders with the two extreme mono-atomic cases and one folder for the alloy of interest. The directories for the alloys contain a second hierarchical level of sub-directories associated with different chemical compositions, and each sub-directory for a chemical composition contains a third hierarchical level with multiple sub-directories, one per atomic configuration. For the binary alloys, the convention used to name the directories associated with different chemical compositions are AXBY, where A and B refer to the constituents, whereas X and Y are positive integers that represent the number of A-atoms and B-atoms in the atomic structure, respectively. The values of X and Y sum up to 128, which is the total number of sites in the Bravais lattice. Similarly, the convention used to name the directories for ternary alloys is AXBYCZ, where A, B, and C refer to the constituents, and X, Y, and Z are positive integers that represent the number of atoms for each constituent and their values still sum up to 128. Inside each directory associated with a chemical composition, 100 sub-directories are generated using the convention 'case-N' to name the directories, where N is a positive integer that spans all the values from 1 through 100, extremes included.

Each subdirectory with name "case-N" contains the following files:

- INCAR: input file that contains various parameters and settings for controlling the behavior of the electronic structure calculations
- KPOINTS: input file that specifies the Bloch vectors (k points) used to sample the Brillouin zone
- 0.POSCAR: input file that defines the atomic structure of a system

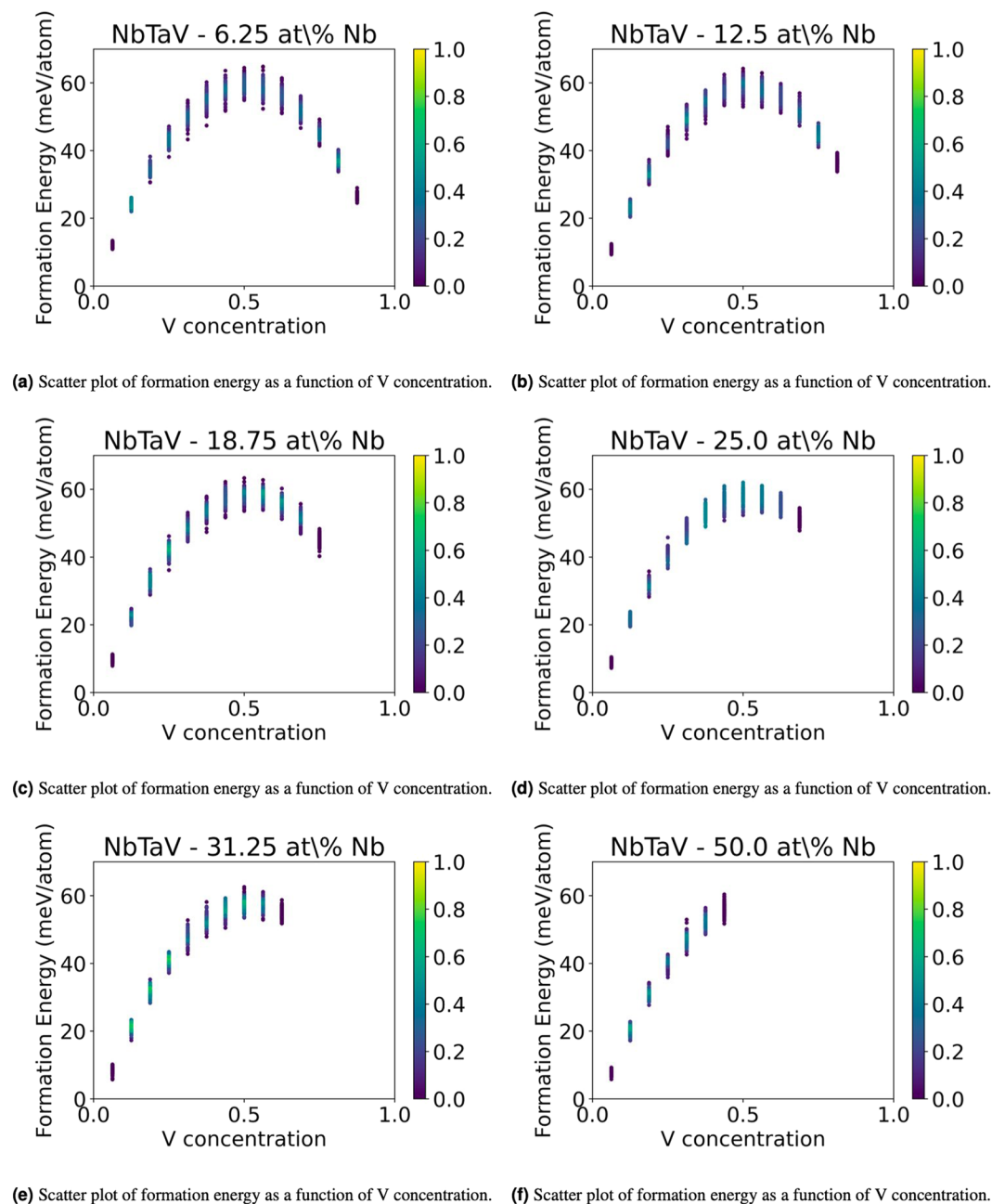


Fig. 3 2D plots of the formation energy of solid solution NbTaV alloys for fixed values of concentrations of Nb.

- 0 . CONTCAR: output file that provides the atomic positions and cell parameters after the first geometry optimization has been run with the precision variable set to “PREC=Low” in the INCAR file
- 0 . OUTCAR: output file that contains detailed information about the progress of a calculation after the first geometry optimization has been run with the precision variable set to “PREC=Low” in the INCAR file
- rlx1 . out: file with diagnostic information about the execution of the first geometry optimization with precision variable set to “PREC=Low” in the INCAR file
- POSCAR: input file that defines the atomic structure of a system after the first geometry optimization has been run at low precision. This represents the input for the second geometry optimization run with the precision variable set to “PREC=Normal” in the INCAR file
- CONTCAR: output file that provides the atomic positions and cell parameters after the second geometry optimization has been run with the precision variable set to “PREC=Normal” in the INCAR file
- OUTCAR: output file that contains detailed information about the progress of a calculation after the second geometry optimization has been run with the precision variable set to “PREC=Normal” in the INCAR file
- rlx2 . out: file with diagnostic information about the execution of the second geometry optimization with precision variable set to “PREC=Normal” in the INCAR file

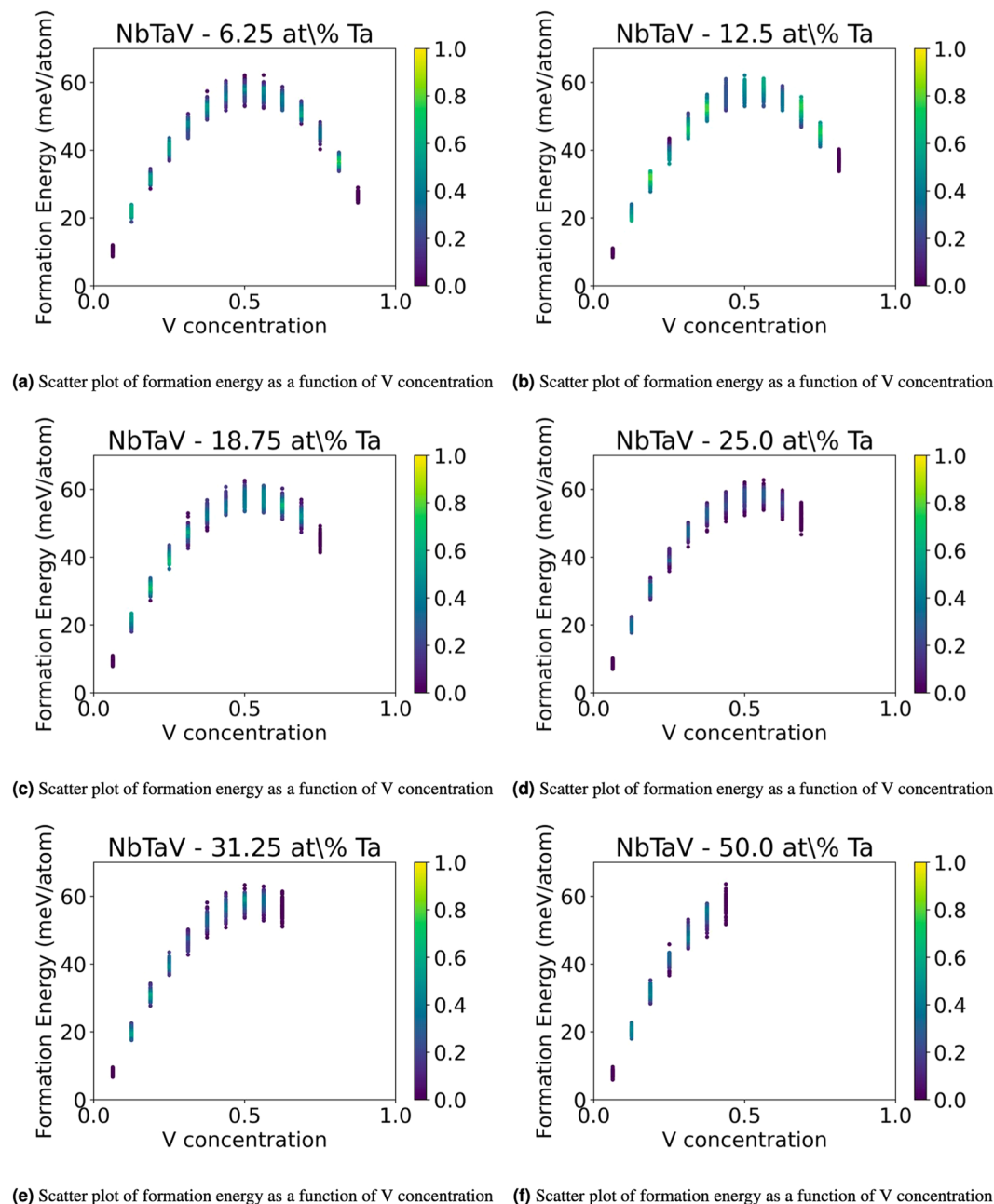


Fig. 4 2D plots of the formation energy of solid solution NbTaV alloys for fixed values of concentrations of Ta.

- `vaspout.h5`: hierarchical HDF5 file containing the inputs and outputs of a VASP calculation. To analyze the data in this file we recommend using `py4vasp`. This file is only produced if the VASP version used is compiled with HDF5 support
- `vasprun.xml`: contains similar information to OUTCAR, but in an xml format.

For more information about the content of the output files we refer the reader to the website https://www.vasp.at/wiki/index.php/Category:Output_files.

Calculation of the formation energy for each atomic configuration. For each atomic configuration, we calculated the formation energy at the ground state (0 Kelvin). At the ground state, the total energy E of an alloy can be written as

$$E = \sum_{i=1}^{N_{\text{elem}}} c_i E_i + \Delta E_{\text{form}}, \quad (1)$$

where N_{elem} is the total number of distinct elements in the system, c_i is the molar fraction of each element i , E_i is the molar energy of each element i , and ΔE_{form} is the formation energy. We calculated the formation energy for

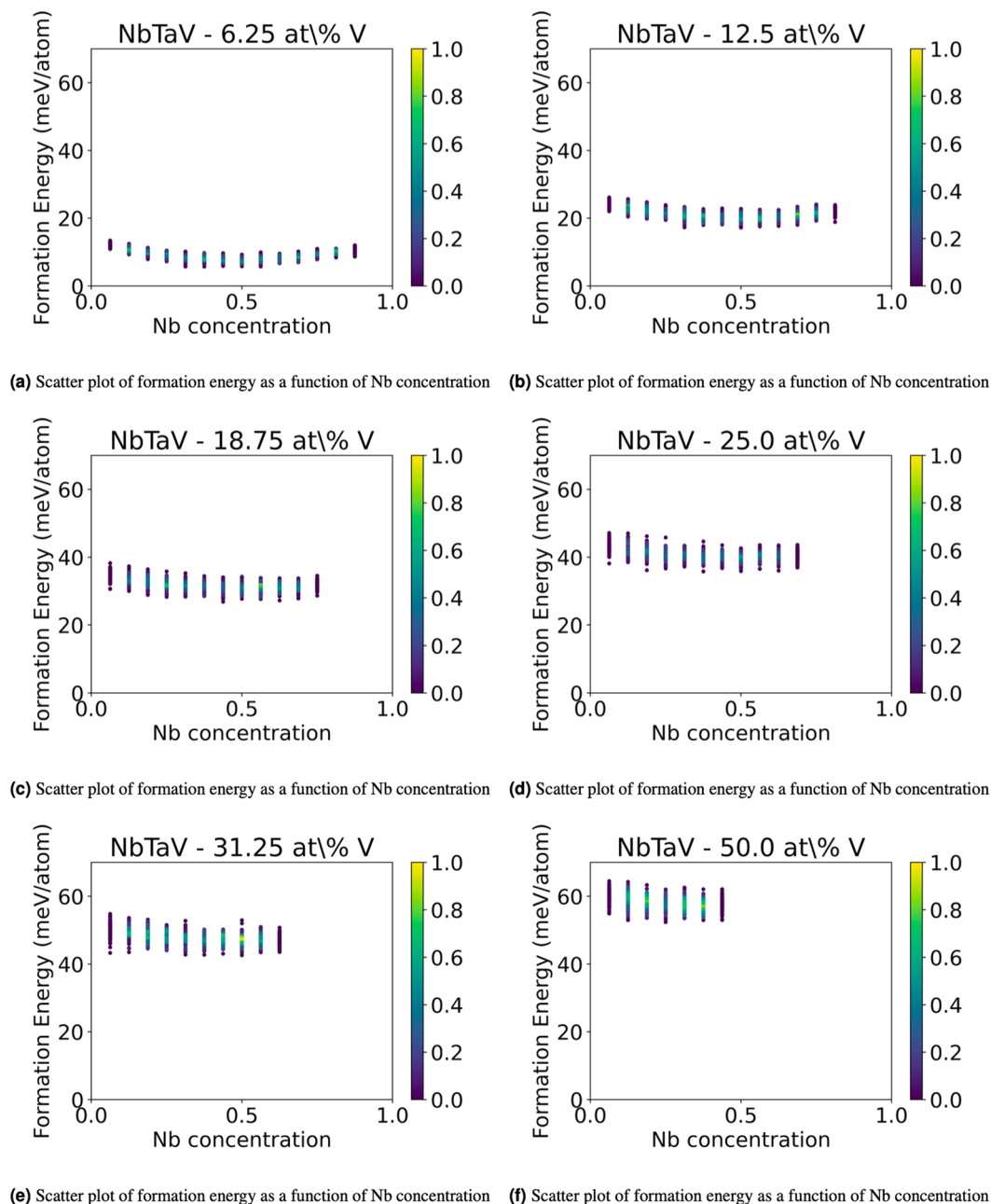


Fig. 5 2D plots of the formation energy of solid solution NbTaV alloys for fixed values of concentrations of V.

each sample by subtracting the internal energy from the total energy computed with VASP. The histograms of the observed formation energies are shown in Figs. 1 and 2. As calculations are performed for a discrete number of concentrations, for each concentration, the energies of the sampled configurations will be centered around the average formation energy for this concentration. This results in multi-peaked energy histograms in Figs. 1 and 2.

Correlation between lattice distortion and mechanical strength of solid solution alloys. In a single-phase HEAs, different elemental atoms with different atomic sizes and number of valence electrons are randomly mixed in one crystallographic lattice, which leads to severe lattice distortion in the atomic structures of HEAs^{42,43}. Lattice distortion has been recognized to have a significant role in strength⁴⁴, phase stability^{45,46}, sluggish diffusion^{47,48}, electrical^{49,50}, and thermal conductivities⁵⁰ of HEAs. Therefore, in this work, we calculated the root mean-squared displacement (RMSD) for each atomic configuration to obtain quantitative understanding on the lattice distortion.

Calculation of the root mean-squared displacement (RMSD) for each atomic configuration. Let us with $X_{\text{ideal}} \in \mathbb{R}^{N_{\text{atoms}} \times 3}$ the set of coordinates of the atoms' nuclei on the ideal BCC Bravais lattice, with $S \in \mathbb{R}^{3 \times 3}$

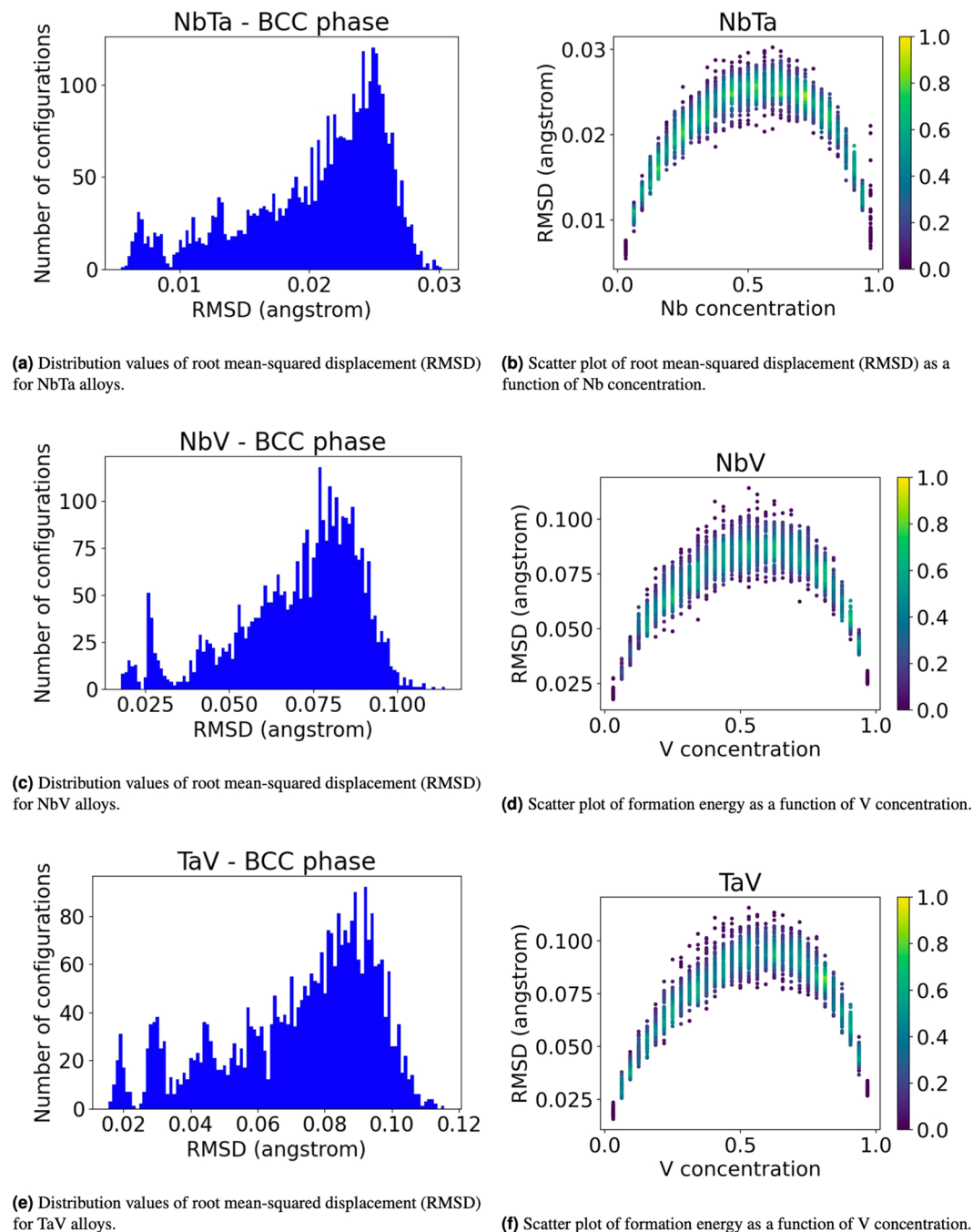


Fig. 6 Root mean-squared displacement RMSD of solid solution NbTa, NbV, and TaV binary alloys with BCC structure.

the deformation tensor computed by the VASP code on each atomic configuration, and with $X_{\text{optimized}} \in \mathbb{R}^{N_{\text{atoms}} \times 3}$ the fully optimized geometry of an atomic configuration. We compute $X_{\text{deformed}} \in \mathbb{R}^{N_{\text{atoms}} \times 3}$ as follows:

$$X_{\text{deformed}}^T = S X_{\text{ideal}}^T, \quad (2)$$

where the superscript T denotes the transpose. We denote with $\mathbf{x}_{\text{optimized}}^i$ the i th row of the tensor X_{ideal} and with $\mathbf{x}_{\text{deformed}}^i$ the i th row of the tensor X_{deformed} , with i being an integer that ranges between 1 and N_{atoms} . The root mean-squared displacement (RMSD) is computed as follows:

$$RMSD = \frac{1}{N_{\text{atoms}}} \sum_{i=1}^{N_{\text{atoms}}} \|\mathbf{x}_{\text{optimized}}^i - \mathbf{x}_{\text{deformed}}^i\|, \quad (3)$$

where $\|\cdot\|$ is the Euclidean norm of a vector.

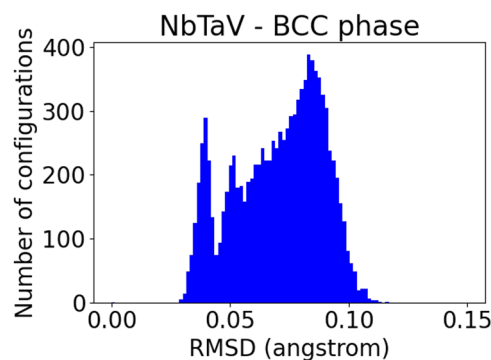


Fig. 7 Distribution values of root mean-squared displacement (RMSD) for NbTaV alloys.

NbTa_BCC_SolidSolution_128atoms_VASP6 dataset. Figure 1(a) illustrates the distribution of the values of the formation energy and Fig. 1(b) illustrates the scatter plot of the formation energy as a function of the Nb concentration for the binary systems NbTa. The convex shape of the scatter plot shows that the BCC solid solution phase favors mixing of the constituents at 0 Kelvin. Figure 6(a) illustrates the distribution of the values of the RMSD and Fig. 6(b) illustrates the scatter plot of the RMSD as a function of the Nb concentration for the binary systems NbTa. Figure 6(b) shows that the maximum RMSD in the binary system NbTa is at approximately 60 % concentration of Nb.

NbV_BCC_SolidSolution_128atoms_VASP6 dataset. The distribution of the values of the formation energy and the scatterplot of the formation energy as a function of the V concentration for the binary systems NbV are shown in Fig. 1(c),(d), respectively. In this case, the scatter plot is concave, thereby indicating a phase separation between the constituents at 0 Kelvin. Figure 6(c) illustrates the distribution of the values of the RMSD and Fig. 6(d) illustrates the scatter plot of the RMSD as a function of the V concentration for the binary systems NbV. Figure 6(d) shows that the maximum RMSD in NbV binary is at approximately 60 % concentration of V. The absolute value is greater than that in the binary system NbTa, suggesting that the random mixing induced lattice distortion between Nb and V greater than the distortion induced in NbTa.

TaV_BCC_SolidSolution_128atoms_VASP6 dataset. Figure 1(e),(f) illustrate the distribution of the values of the formation energy and the scatter plot of the formation energy as a function of the V concentration for the binary systems TaV, respectively. Also for this system, the scatter plot is concave, thereby indicating a phase separation between the constituents at 0 Kelvin. Figure 6(e) illustrates the distribution of the values of the RMSD and Fig. 6(f) illustrates the scatter plot of the RMSD as a function of the V concentration for the binary systems TaV. Figure 6(f) shows that the maximum RMSD in the binary system TaV is at approximately 60 % concentration of V. The absolute value is the greatest among the Nb-Ta, Nb-V and TaV systems, suggesting that the random mixing induced the greatest distortion in the binary system TaV.

NbTaV_BCC_SolidSolution_128atoms_VASP6 dataset. Figure 2 illustrates the distribution of the values of the formation energy, whereas Figs. 3, 4, and 5 illustrate scatter plots of the formation energy for the ternary alloys NbTaV at fixed values of concentrations of Nb, Ta, and V, respectively. For low concentrations of Nb, the plots in Fig. 3 have a similar shape to the ones of the binary alloy TaV in Fig. 1(f). Similarly, for low concentrations of Ta, the plots in Fig. 4 are very similar to the ones of the binary alloy NbV in Fig. 1(d). Finally, for low concentrations of V (say 6.25 at% V), the plots in Fig. 5 show a convex shape of the formation energy as a function of the chemical concentration of V, thereby showing a very similar trend to the one of the formation energy for the binary alloy NbTa in Fig. 1(b). These trends are reasonable to expect, because small additions of a third new constituent to a binary alloy are not supposed to abruptly change the trend of the formation energy as a function of the chemical composition. Fig. 7 illustrates the distribution of the values of the RMSD, whereas Figs. 8, 9, and 10 illustrate scatter plots of the RMSD for the ternary alloys NbTaV at fixed values of concentrations of Nb, Ta, and V, respectively. For all sampled concentrations of Nb, the RMSD reaches a maximum for chemical compositions that are rich in V. The magnitude of RMSD does not show strong dependency on Nb concentration. The random mixing between Ta and V induced a RMSD in the ternary that is not significantly different from the one induced in the binary TaV. A similar trend is also observed for the NbV mixing at different concentrations of V. However, for the NbTa mixing, the RMSD significantly increases with the increasing concentrations of V. These results suggest that the mixing of V with the other two elements is the key factor that leads to larger RMSD and, thus, leads to increased strength of the ternary systems NbTaV.

Artefact Description. The datasets NbTa_BCC_SolidSolution_128atoms_VASP6, NbV_BCC_SolidSolution_128atoms_VASP6, and TaV_BCC_SolidSolution_128atoms_VASP6 contain each 31 directories associated with different chemical compositions, and each directory associated with a chemical composition contains 100 sub-directories, one per atomic configuration. This amounts to a total of 3,100 sub-directories. The sub-directories with name “case-*N*”, where *N* ranges between 1 and 10 (extremes included) contain all the

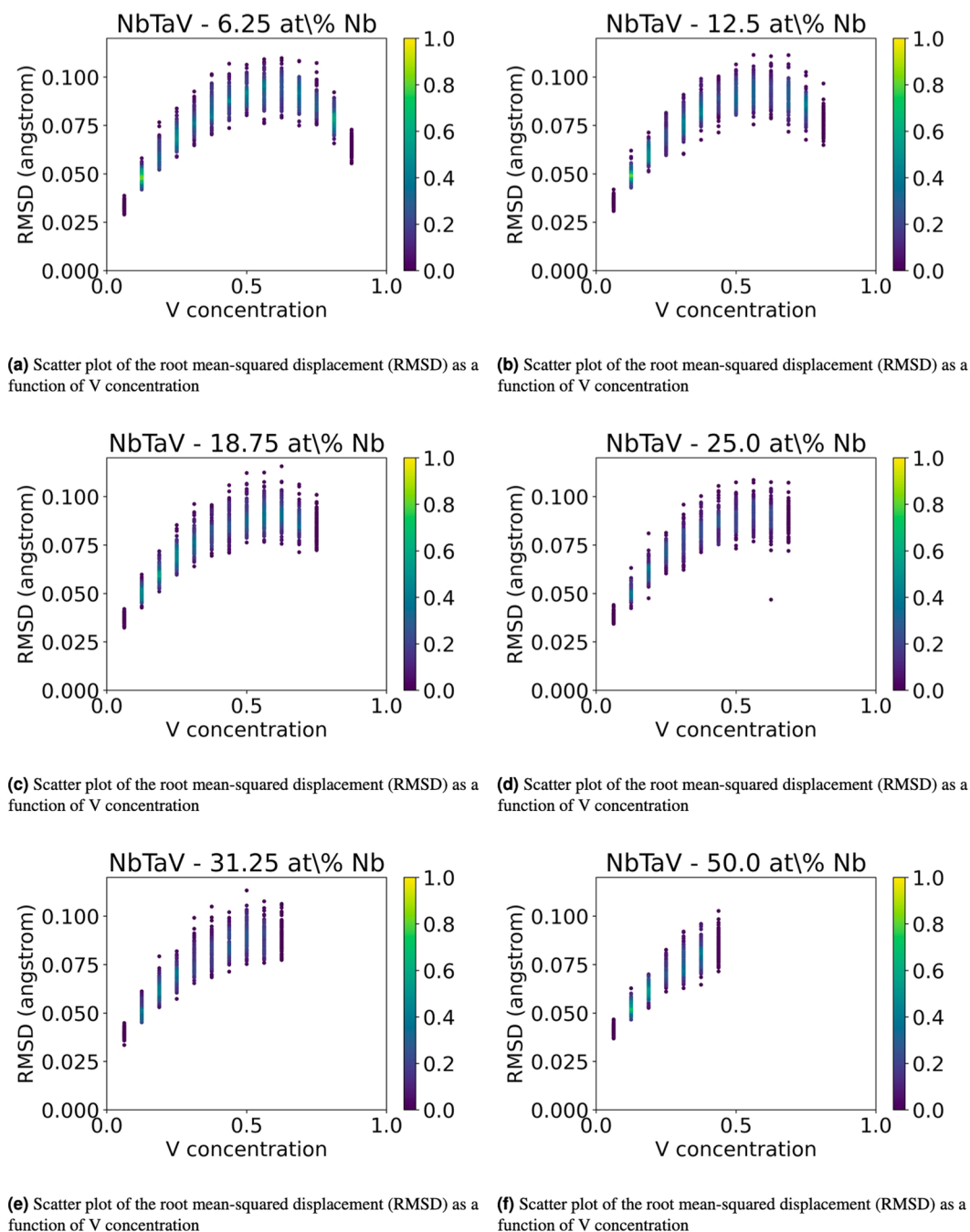


Fig. 8 2D plots of the root mean-squared displacement (RMSD) of solid solution NbTaV alloys for fixed values of concentrations of Nb.

files listed in Section except `vaspout.h5`. Sub-directories with name “case-*N*”, where *N* ranges between 41 and 60 (extremes included), contain a duplicate copy of the files listed above except for `KPOINTS`. The names of the duplicate files end with “-bis”, and correspond to a second VASP calculation that has converged to a different optimized geometry. The differences in these duplicate calculations are for calculations performed on the two different computing resources described above including different hardware architectures and compilers. Thus these differences arise from slight numerical differences in the calculations and the difference in the relaxed atomic positions is of the order of only 0.01% of the supercell size and the energy difference of the order of a few μeV per atom. The dataset `NbTaV_BCC_SolidSolution_128atoms_VASP6` contain 105 directories associated with different chemical composition, and each directory associated with a chemical composition contains 100 sub-directories, one per atomic configuration. This amounts to a total of 10,500 sub-directories.

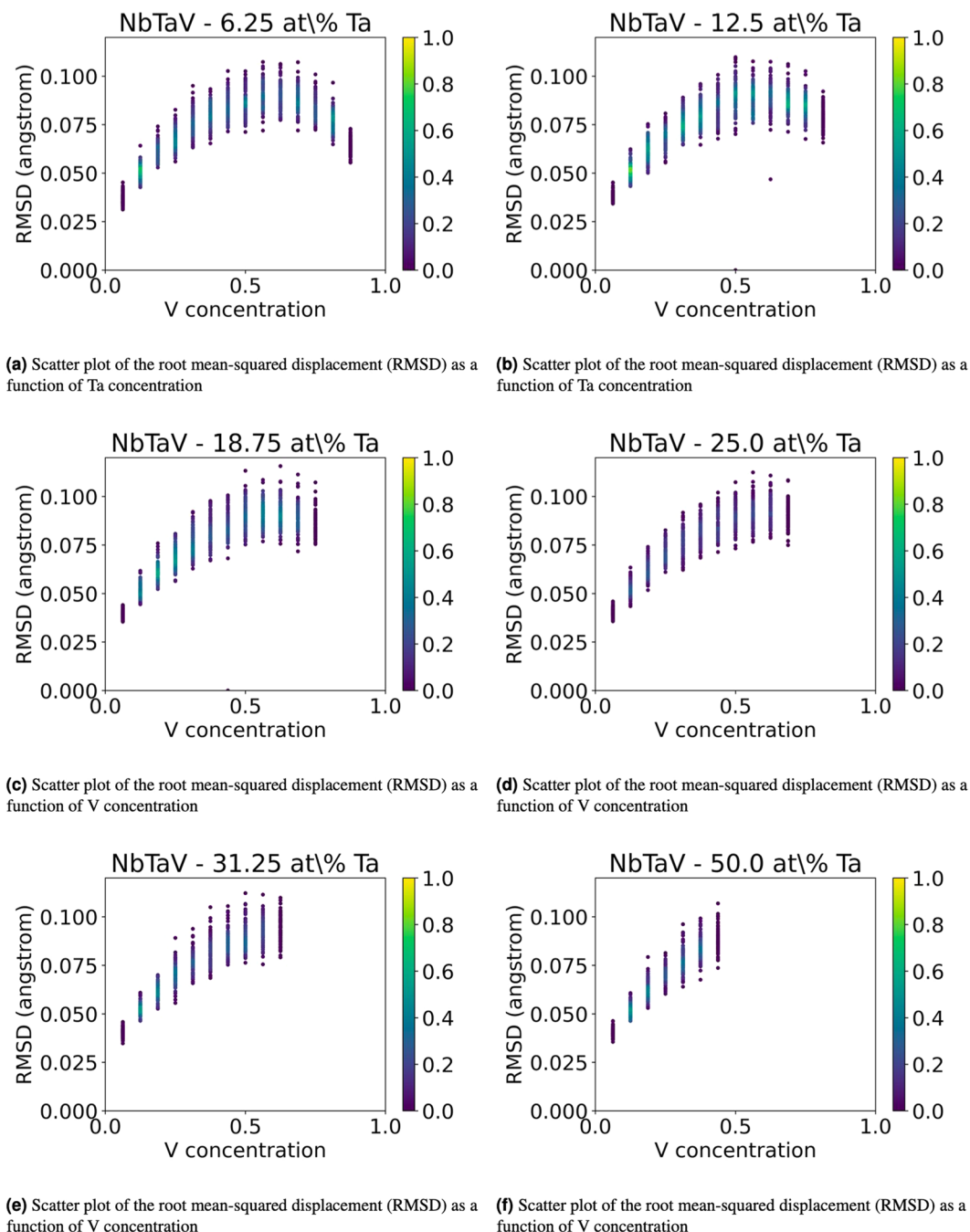


Fig. 9 2D plots of the root mean-squared displacement (RMSD) of solid solution NbTaV alloys for fixed values of concentrations of Ta.

Data Records

The open-source datasets NbTa_BCC_SolidSolution_128atoms_VASP6²⁹, NbV_BCC_SolidSolution_128atoms_VASP6³⁰, TaV_BCC_SolidSolution_128atoms_VASP6³¹, and NbTaV_BCC_SolidSolution_128atoms_VASP6³² are stored by the OLCF Data Constellation Facility. The datasets can be downloaded using the Globus data transfer service, as indicated by the instructions provided at the following website <https://docs.olcf.ornl.gov/data/index.html#data-transferring-data>.

Alternatively, the datasets are also available on Figshare and can be downloaded from the following websites:

- https://figshare.com/articles/dataset/NbTa_BCC_SolidSolution_128atoms_VASP6_-_DOI_10_13139_OLCF_2222906/25810894
- https://figshare.com/articles/dataset/NbV_BCC_SolidSolution_128atoms_VASP6_-_DOI_10_13139_OLCF_2228839/25806814

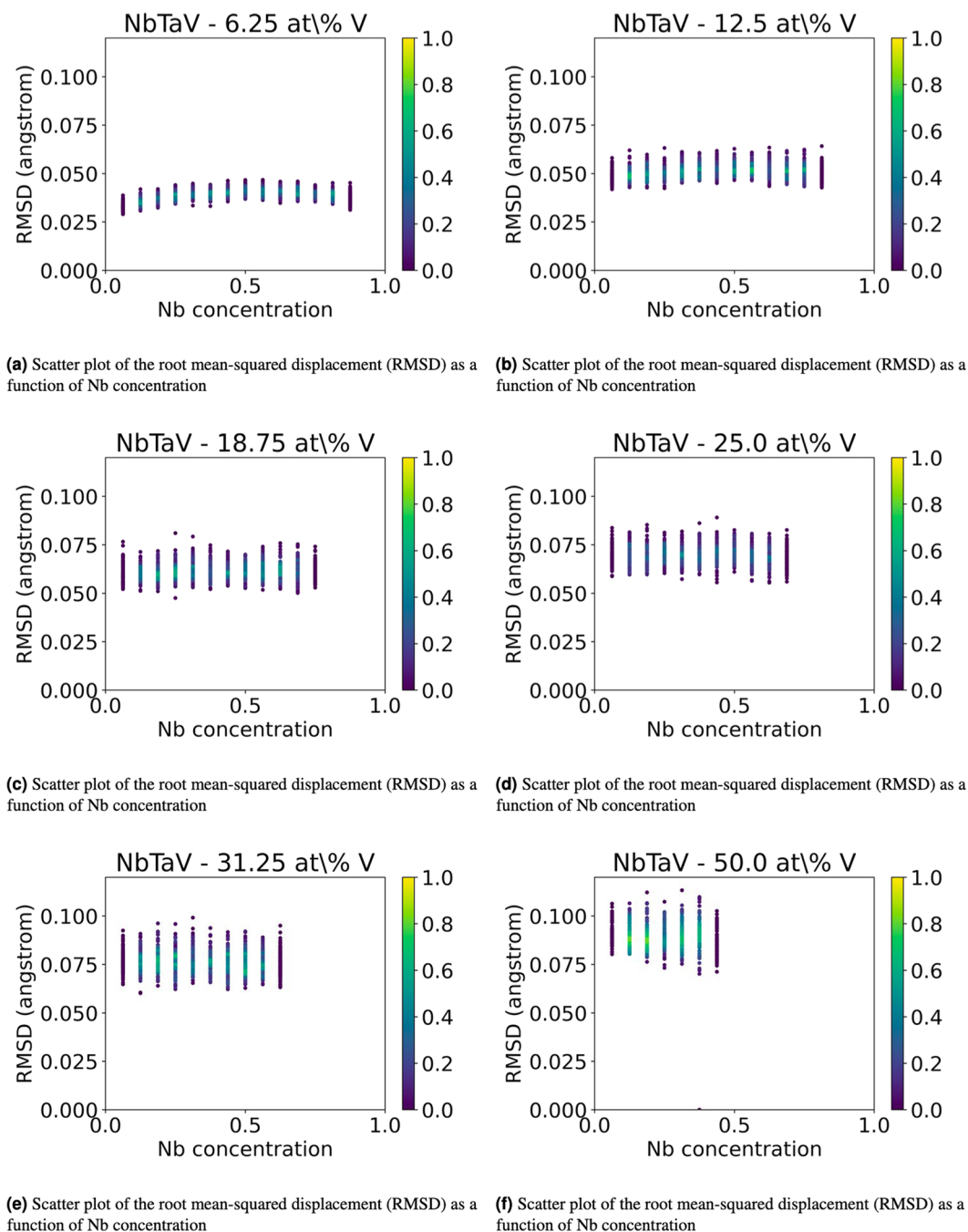


Fig. 10 2D plots of the root mean-squared displacement (RMSD) of solid solution NbTaV alloys for fixed values of concentrations of V.

- https://figshare.com/articles/dataset/TaV_BCC_SolidSolution_128atoms_VASP6_-_DOI_10_13139_OLCF_2222910/25805131
- https://figshare.com/articles/dataset/NbTaV_BCC_SolidSolution_128atoms_VASP6_-_Part_1_-_DOI_10_13139_OLCF_2217644/25824178
- https://figshare.com/articles/dataset/NbTaV_BCC_SolidSolution_128atoms_VASP6_-_Part_2_-_DOI_10_13139_OLCF_2217644/25838581

Technical Validation

Importance of large Bravais lattices to accurately model the configurational entropy in solid solutions. The authors of⁵¹ calculated phase diagrams of NbV, TaV, and NbTa alloys by combining the total energies of 40-50 configurations for each system (obtained via DFT calculations run with the VASP software) with the cluster expansion and Monte Carlo techniques. For this study, a BCC supercell with 12 atoms was used

for the DFT calculations. Although this study represents a seminal work for solid solution alloys in the Nb-Ta-V chemical space, the size of the BCC supercell used and the total number of atomic configurations considered are not sufficiently large to make meaningful conclusions about the solid solution phases of NbV, TaV, and NbTa alloys. In fact, using a BCC supercell with only 12 atoms and running 40–50 configurations for each system over the entire compositional range does not capture the effect of the configurational entropy on the formation energy of the systems, thereby precluding meaningful conclusions about the stability (or instability) of solid solution phases at certain chemical compositions and temperatures. This is corroborated also by comparing the values of the formation energy at 0 Kelvin obtained from our DFT calculations for solid solution binary systems NbTa, NbV, and TaV in Figure 1 of this manuscript and the DFT values mixed with the cluster expansion (CE) predictions in Figures 7, 9, and 11 of⁵¹. Even though we must be mindful in comparing our results that represent only DFT values with the ones displayed in Figures 7, 9, and 11 of⁵¹ which mix DFT values with CE predictions, a qualitative comparison shows that the DFT values from our calculations are less scattered and more concentrated around the parabolic curve. This indicates that our calculations better capture the effect of atomic disorder on the stability/instability of the solid solution phase, depending on whether the parabola is convex or concave, whereas the formation energy values in Figures 7, 9, and 11 of⁵¹ are dispersed over a broad range of values because BCC structures with only 12 atoms can only describe ordered (e.g., intermetallic) phases.

Usage Notes

The code to prepare the inputs of the VASP calculations and performing post-processing statistical analysis of the dataset is open-source and available at the ORNL-GitHub repository https://github.com/ORNL/ScalableWorkflow_VASP_Calculations. The code contains the following Python script:

- `generate_vasp_atomic_configurations_binary.py`: generates the hierarchy of directories (with one directory per chemical composition and multiple subdirectories names `case-N`, with N between 1 and 100, for each chemical composition of the binary alloy)
- `generate_vasp_atomic_configurations_ternary.py`: generates the hierarchy of directories (with one directory per chemical composition and multiple subdirectories names `case-N`, with N between 1 and 100, for each chemical composition of the ternary alloy)
- `compute_formation_energy.py`: computes the formation energy for each fully optimized atomic structure by subtracting the linear term of the total energy
- `plot_formation_energy_binary.py`: plots the histogram for the distribution of values of the formation energy for the binary alloys and generates the scatterplot of the formation energies for each atomic structure as a function of the chemical concentration of one of the two constituents of the binary alloys
- `plot2D_sliced_formation_energy_ternary_ase_objects`: plots the histogram for the distribution of values of the formation energy for the binary alloys and generates the scatterplots of the formation energies for each atomic structure by fixing the chemical concentration of one constituent, and plotting the formation energy values as a function of the chemical concentration of another constituent of the ternary alloys

Code availability

The code to prepare the inputs of the VASP calculations and performing post-processing statistical analysis of the dataset is open-source and available at the ORNL-GitHub repository https://github.com/ORNL/ScalableWorkflow_VASP_Calculations.

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Author contributions

M.L.P. created the input files for each disordered atomic configuration, curated the dataset once its collection was completed for its public release, analyzed the dataset, and drafted the main portion of the manuscript. G.S. ran the VASP calculations on the NERSC-Perlmutter system for binary solid solution NbTa, NbV, and TaV solid solution alloys with cases 21-40 for each chemical composition, and ran all the cases for the ternary solid solution NbTaV alloys, participated in writing the manuscript. M.E. ran the VASP calculations on the OLCF-Summit system with cases 41-100 for each chemical composition of the binary solid solution Nb-Ta, Nb-V, and Ta-V alloys. J.Y.C. created efficient bash scripts to effectively run large ensembles of VASP calculations on NERSC-Perlmutter and OLCF-Summit, and helped diagnose the runtime errors obtained during the execution of the calculations. J.Y. ran the VASP calculations on the OLCF-Summit system with cases 1-10 for each chemical composition of the binary solid solution NbTa, NbV, and TaV alloys. Y.Y. chose the alloys to study based on her experimental observations obtained through her laboratory experiments and provided meaningful interpretation of the output of the simulations.

Competing interests

The authors declare no competing interests.

Additional information

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